Clustering decision audit

Method	Space	Best k	Criterion used	Criterion value	Stability (Jaccard)	Fisher p
Ward.D2	Full 17D	1	Gap (1−SE)	1.212	−	−
K−means	PCA 4D	2	Gap; PCs−>stability	1.091	0.929	0.629
GMM	PCA 4D	2	ICL; PCs−>stability	−1702.6	0.682	1.000
FCM	Full 17D (m=1.5)	2	Xie−Beni (lower; k>=2)	16.245	0.924	0.475
UMAP+Km	UMAP 2D (10/0.15)	2	Silh.; params−>stability (k>=2)	0.434	0.774	1.000

Criteria are method−specific and cannot be compared across rows. Higher values are preferred for Gap, ICL and silhouette; lower values are preferred for Xie−Beni. S